

The University of the South Pacific

School of Information Technology, Engineering, Mathematics & Physics

CS219: Cloud Computing

Assignment 1 - 2026

Total Marks: 20 (10% of CW)

Due Date: Thursday 3rd September 2026 at 11:55pm

This assessment covers the following course learning outcome:

- I. CLO 1 – Explain the capabilities and limitations of cloud computing.
- II. CLO 2 - Explain the roles and responsibilities of different sub-systems in cloud architecture.

Part A

Scenario

Mere, a recent graduate with a Bachelor's degree in Computer Science and Information Systems from The University of the South Pacific, has noticed that many small businesses in Fiji and other Pacific Island countries are increasingly using social media and online platforms to reach customers. However, some small businesses still rely heavily on manual processes for managing customer orders, inventory, payments, and communication.

After discussing these challenges with several local business owners, Mere identifies an opportunity to establish a small **online marketplace that connects local producers and small businesses with customers**. The platform could allow customers to browse products, place orders, make online payments, and receive updates about their purchases.

Rather than purchasing and maintaining her own physical servers, Mere is considering using cloud **computing and Software-as-a-Service (SaaS) solutions** to develop and operate the business. This approach could allow her to start with relatively low infrastructure costs and scale the business as the number of customers and participating businesses increases.

Before proceeding, Mere needs to determine whether a cloud-based business is feasible in the Pacific context and understand the benefits, limitations, costs, infrastructure requirements, and stakeholders involved.

The following questions require you to evaluate Mere's proposed business and demonstrate how cloud technology and the SaaS delivery model could be applied to establish and operate the online business.

Question 1 (5 marks)

Evaluate the feasibility of Mere establishing and operating an online marketplace using cloud computing. In your response, discuss both the potential benefits and challenges of adopting cloud technology. Your discussion should address the following:

i. Definition of Cloud Computing

Define cloud computing and explain its main characteristics.

ii. Advantages of Cloud Computing

Discuss the key advantages of using cloud computing for Mere's proposed online marketplace.

iii. Disadvantages and Limitations

Discuss the potential disadvantages, risks, and limitations Mere may face when using cloud computing.

iv. Considerations in the Pacific Island Context

Discuss factors that may affect the feasibility of Mere's cloud-based business in Fiji and other Pacific Island countries. Consider issues such as internet connectivity, cost, digital infrastructure, cybersecurity, availability of technical expertise, and reliability of cloud services.

Question 2 (5 marks)

Using Mere's online marketplace as an example, explain how she could establish and operate the business using the Software-as-a-Service (SaaS) cloud delivery model. Your response should identify the key stakeholders involved and explain how the different components of the cloud-based solution interact. Your discussion should address the following:

i. Software-as-a-Service (SaaS) Model

Explain the SaaS cloud delivery model and describe how it differs from traditional software installation and management.

ii. Proposed Online Business

Describe how Mere's online marketplace could operate using SaaS. Explain the main services and functions that would be provided to customers and participating local businesses.

iii. Cloud Architecture

Provide a **cloud architecture diagram** showing the major components of the proposed solution and how they interact. The diagram should identify relevant stakeholders and services, such as customers, Mere/business administrators, SaaS providers, cloud infrastructure, databases, payment services, and other relevant third-party services. Accompany the diagram with a detailed explanation of how the components communicate and support the business.

iv. Advantages, Disadvantages and Future Considerations

Discuss the advantages and disadvantages of using SaaS for Mere's business. Suggest possible future considerations, such as business growth, scalability, data security, integration with other services, mobile applications, additional payment options, and expansion to other Pacific Island markets.

Where appropriate, support your discussion with relevant examples, diagrams, academic sources, and references.

Part B

Question 3 (10 marks)

Use Oracle Virtual Box hypervisor to configure the following requirements:

- i. Create a new Virtual Machine (VM) using the Windows Server 2012 Identical Storage Image (ISO) Operating system. RAM Size: 1024MB or higher, Storage: 100GB or Higher, Network Adapter [1 x Host Only Adapter, 1 x Bridged Adapter], Virtual Machine Name: SRVMA1.
- ii. Assign Class A static IPv4 address on LAN. For instance, SRVMA1 10.0.0.10 255.0.0.0 & WAN should receive an IP Address automatically from the Internet. Change Computer Name to SXXXXXXXSRVMA1. Set the administrator password: P@\$w0rd12345
- iii. Configure a Primary Domain as SXXXXXXX.local and create two user accounts including you and your partner details. Use naming convention: <firstname.surname>.
- iv. Create another Virtual Machine and install a Windows 10 OS. RAM Size: 1024MB, Storage: 80GB or Higher, Network Adapter [1 x Host Only Adapter, 1 x Bridged Adapter], Virtual Machine Name & Computer Name: CLVMA1. Assign appropriate IP addressing on LAN for connectivity to the Server created in Part a above. Join the CLVMA1 to the domain SXXXXXXX.local.
- v. Create a folder on SRVMA1 called "CLOUD". Allow Full Control to your account & Read Access to your partner's account. Ensure to give appropriate permissions and apply security authentications.
- vi. Configure Virtual Private Network to securely connect from the untrusted network (Internet) to a secured environment. Note: You should receive IP address assigned for VPN connections and access the CLOUD file service created in e above from the CLVMA1.
- vii. Prepare a **user guide** with explanations and screenshots to document your steps from i – vi above and include it in the Assignment 1 report.

Assignment Rules

This is an **individual assignment**. You are expected to work on your assignment alone however, you are welcome to discuss your ideas with your colleagues and tutors.

You have to submit your work before the deadline mentioned above. Lab computers are scarce, so the trick is to start EARLY. Also, when you finish don't hold on to your assignment till the last moment. Submit as soon as you are satisfied with it.

If you think that you will not be able to submit the assignment on time due to circumstances which are beyond your control, then you will need to seek approval from the course coordinator prior to the due date. This means you have to ask for an extension, explaining the reasons, before the deadline, and not after. A late submission will be penalized by 20% for each 24 hour period it is late. Where possible, it is better to hand in your work early and get credit for partial work than handing in late. A partial work may earn more points than an assignment which is submitted late.

What to submit:

You will have to submit only one file in the **Assignment 1 DropBox** on Moodle.

A Report write-up answering the questions 1-3. This will be one document that you can create using MS Word. Make sure to save it as a PDF before you submit. Your write-up will have to be written in English. Try to keep your answers short and clear. Excessively verbose answers will be penalized. The write-up should have the following sections:

- Cover page with your name, ID, and course details.
- Table of content
- Introduction – a background to the assignment subject & topic
- Question 1 – Business with Cloud Technology (5 marks)
 - Cloud Technology?
 - Advantages
 - Disadvantages
 - Pacific Island nations
- Question 2 – SaaS Small Local Business (5 marks)
 - SaaS cloud delivery model
 - Small online business description
 - Cloud architecture (diagram & text)
 - Advantages, disadvantages, future considerations (if any)
- Question 3 – Oracle Virtual Box hypervisor (10 marks)
 - User Guide for Question 3
- Conclusion – conclude with the topic knowledge covered in the assignment
- Reference – IEEE format. You should add references to recent literature to back up your solutions.

AI Usage Policy and Plagiarism

No AI Usage: AI tools cannot be used in this assessment task. In this assessment, you must NOT use generative AI to generate any materials or content in relation to the assessment task.

For this and other work in CS219, it is essential that you avoid plagiarism. Not only do you expose yourself to possibly serious disciplinary consequences, but you will also cheat yourself of a proper understanding of the concepts emphasized in the project. You will almost certainly fail the short tests and/or the final which will test your understanding of the project. It is not plagiarism to discuss the assignment with your friends and consider solutions to the problems together. However, it is plagiarism for you to copy all or part of each other's programs. If you find somebody has stolen your assignment and produced it as their own, it will be considered plagiarism. We are using automated tools to assist with the detection of plagiarism. They will highlight any unusual writing that is similar between students. All cases that are flagged as potential plagiarism will be checked by hand. So do not leave your flash drives around. Be careful with them. And make sure you log out of the lab machines when you are finished working with them. Don't copy part of someone solutions, and do not give somebody else an electronic copy of your solution. If you give someone your assignment it is almost guaranteed that part of it will end up in their assignment, no matter whether you ask them not to copy. Any student posting answers for this assignment on Moodle will be considered to be committing plagiarism. Do not submit your answers to any discussion group or mail it to anyone except the lecturers or tutors.

Note: If you need clarifications while doing this assignment, you may post a message on the **Assignment Forum** located on the CS219 Course Page under the Assessments block. Students are advised to read through the responses for postings made by other students before posting their own to avoid repetition of the queries.

Marking Guideline

Total	Abstract & Design	CBOK	Unsatisfactory (U)		Satisfactory (S)		Good (G)		Marks Allocated
	Understanding of ICT Profession								
			i. Unable to identify the current trends and issues in ICT domain. ii. Unable to accumulate the history of the evolution of ICT discipline. iii. Unable to analyse the recent issues whilst solving a specific ICT problem.	I. No risk analysis & mitigation. II. Irrelevant prototyping III. Requirements not finalized (all valid/invalid requirements are dumped in SRS document) IV. Instead of Software development methodology tools such as JIRA or Microsoft Visual Studio team foundation simple editor like MS word is used. V. Incorrect flowchart or UML/ERD diagrams. VI. Inappropriate tools used for UML/ERD diagrams, such as MS word. VII. The diagrams does not reconcile with the approved requirements (from SRS document). VIII. Poor architecture and design makes future updates hard to incorporate. IX. No adequate analysis of algorithm for its efficiency and relevance. X. inappropriate design of algorithms (efficiency, structure and data structures)		i. Poorly identified the current trends and issues in ICT domain ii. Accumulate the history of the evolution of ICT discipline but incomplete and unclear. iii. Poor judgement on the analysis of the recent issues whilst solving a specific ICT problem. i. Properly identified the current trends and issues in ICT domain ii. Accumulate the history of the evolution of ICT discipline clearly and organised manner. iii. Analyse the recent issues with proper justification and logical arguments whilst solving a specific ICT problem.		6	
						I. Analysis of requirements such as risk analysis & mitigation, prototyping, identification of true/false requirements etc. II. Software development methodology tools such as JIRA Microsoft Visual Studio team foundation etc. III. all relevant constraints and requirements are mentioned IV. Correct flowchart or UML/ERD diagrams using appropriate tool V. The diagrams depicts the approved requirements (from SRS document). VI. Appropriate choice of the architecture and design VII. Analysis of algorithm for its efficiency and relevance. VIII. Appropriate design of algorithms (efficiency, structure and data structures)		14	
								20	